VIRTUAL CLOTHES TRY-ON SYSTEM







SUBMITTED BY

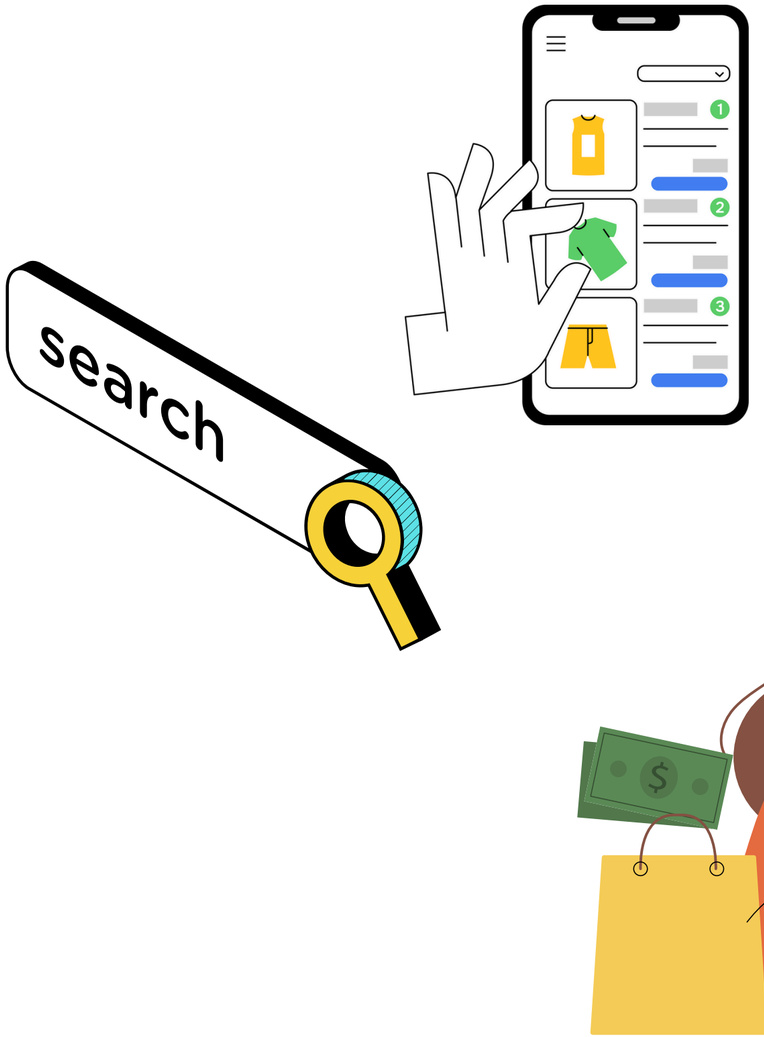
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INTRODUCTION

- Online fashion retail lacks the ability for users to physically try clothes.
- Static images and size charts cause uncertainty.
- Results: higher return rates, reduced trust, and lower satisfaction.
- Solution: Virtual try-on systems simulate how garments look on a user image using AI & computer vision.



SIGNIFICANCE OF VIRTUAL CLOTHES TRY-ON

- **E-Commerce Enhancement:** Enables confident online purchases.
- **Customer Satisfaction:** Reduces size/fit uncertainty.
- **Sustainability:** Cuts return-related waste.
- **Fashion Design:** Digital showcasing without samples.
- **Accessibility:** Inclusive experience for diverse body types.



PROBLEM STATEMENT

Issue:
Customers can’t try garments before purchase → poor fit, high returns, low trust.

Existing Systems:
2D overlays, AR filters, or 3D models — limited realism or high cost.

Challenge:
“How can we design an affordable, scalable, and realistic virtual try-on system using only a photograph and a clothing image?”



OBJECTIVES

- Realistic **clothing visualization** using deep learning.
- Accurate **pose estimation** and body alignment.
- **Warping & blending** to fit garments naturally.
- **Accessible & lightweight** system using open-source tools.
- **E-commerce integration** to enhance online retail.



Clothing Visualization

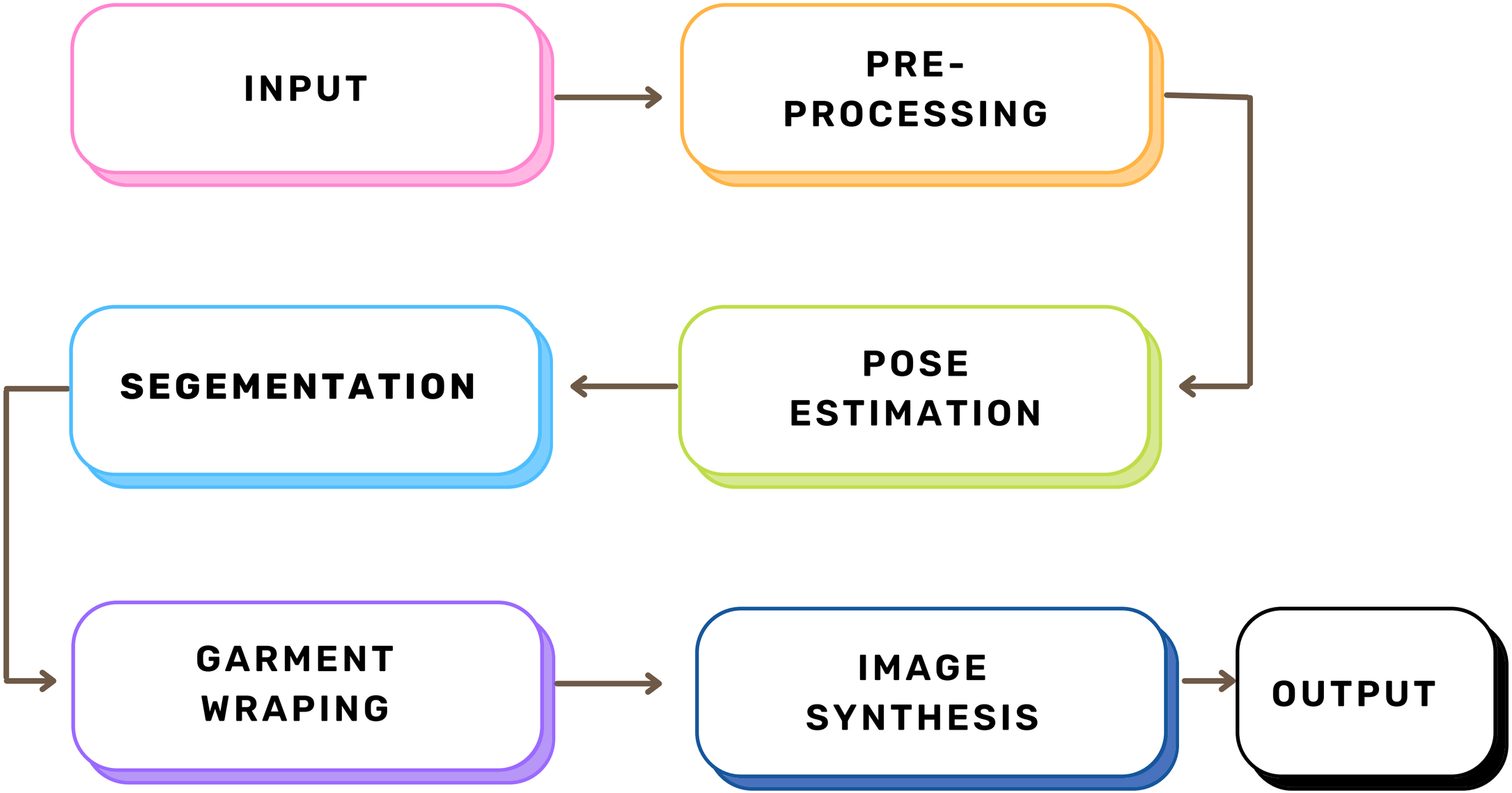
Pose Estimation

Warping & Blending

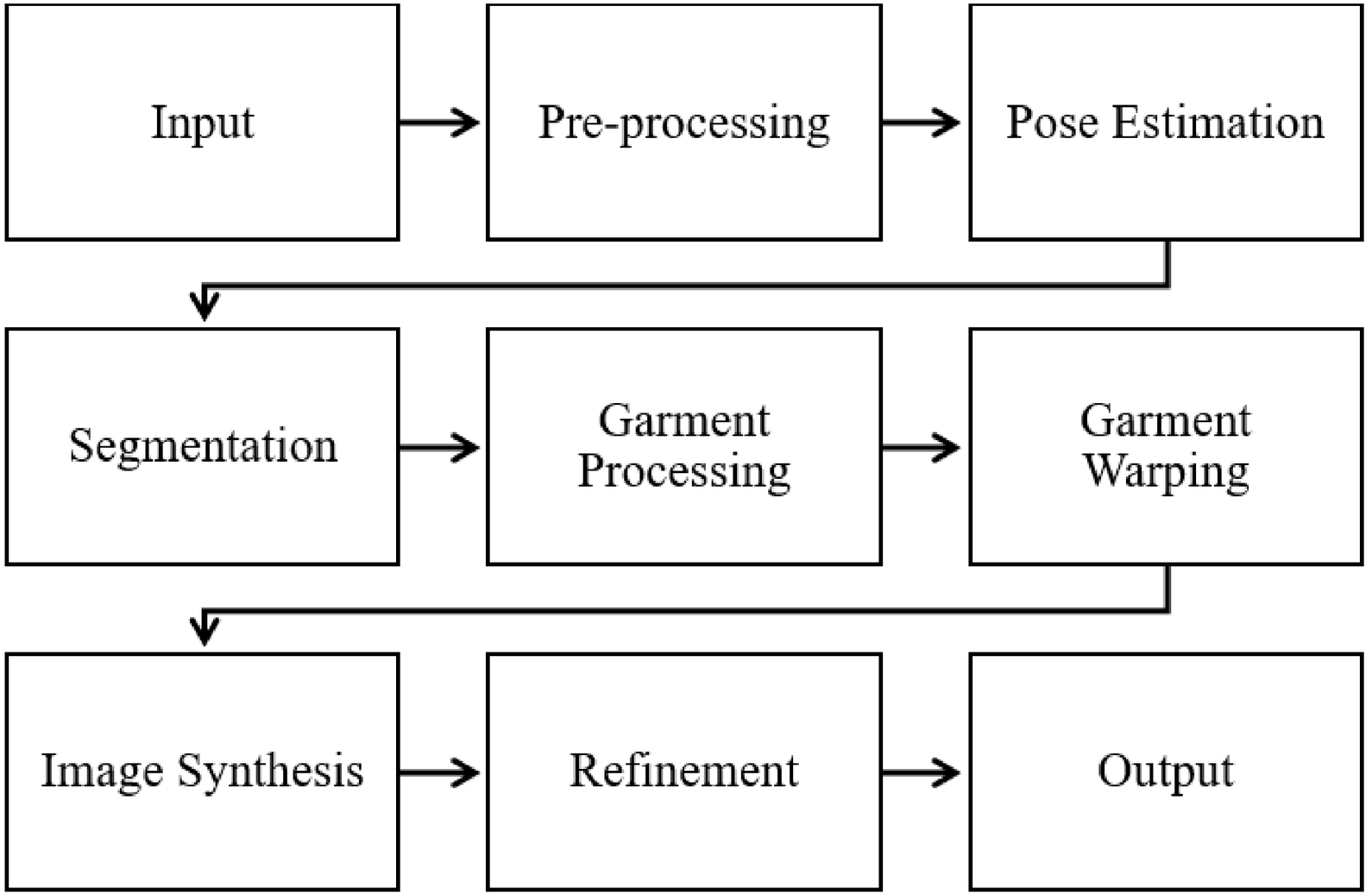
Accessible & Lightweight

E-Commerce Integration

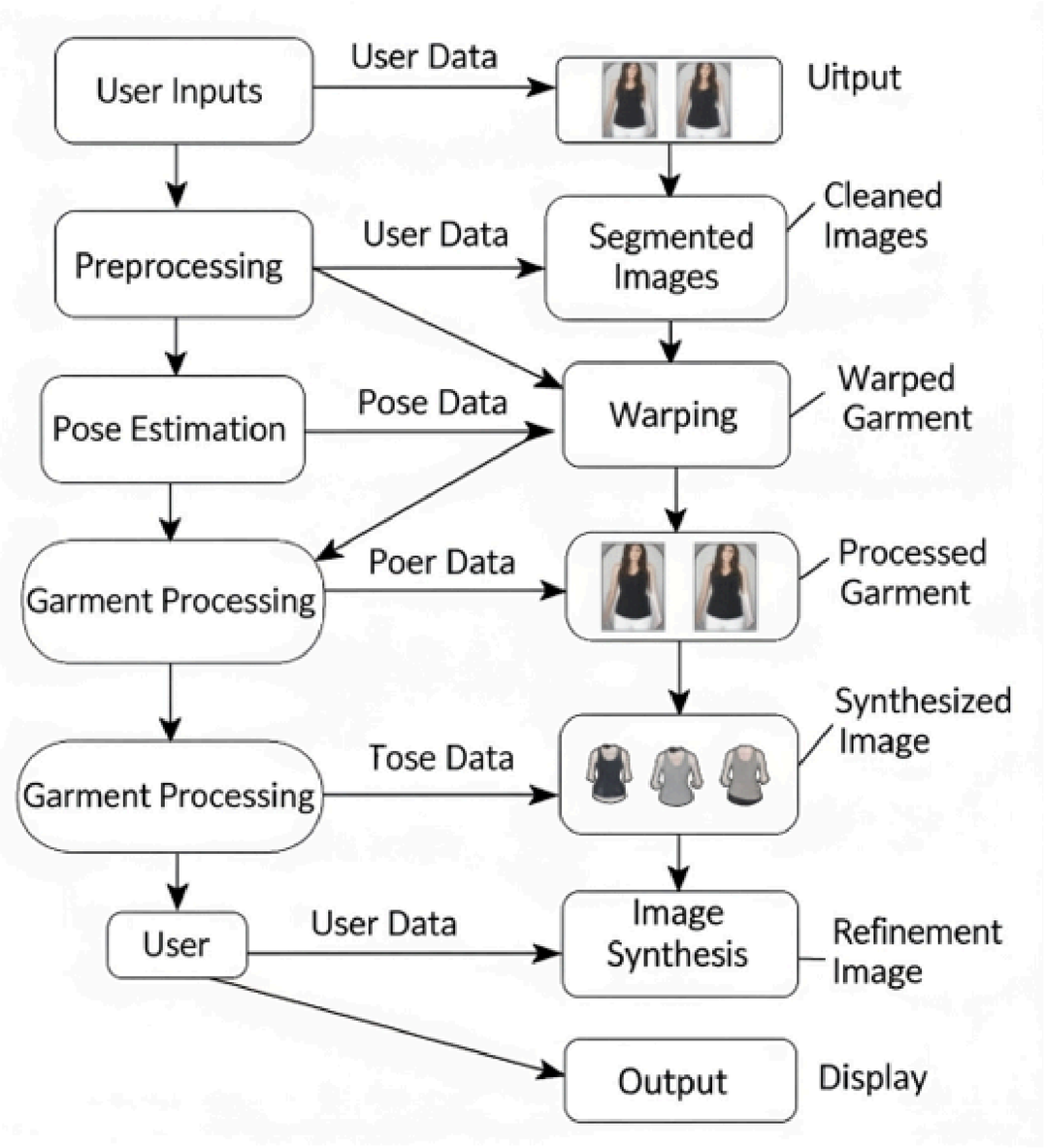
PROPOSED METHODOLOGY



SYSTEM ARCHITECTURE



DFD


User Inputs
User Data
Uitput
Preprocessing
User Data
Segmented Images
Cleaned Images
Pose Estimation
Pose Data
Warping
Warped Garment
Garment Processing
Poer Data
Processed Garment
Garment Processing
Tose Data
Synthesized Image
User
User Data
Image Synthesis
Refinement Image
Output
Display

REQUIREMENT ANALYSIS

HARDWARE

- CPU: i3 or above (i5 recommended)
- RAM: 8 GB (16 GB recommended)
- GPU: GTX 1650 or better
- OS: Windows/Linux/macOS

SOFTWARE

- Python 3.8+
- PyTorch, OpenCV, NumPy, TorchVision
- Jupyter/PyCharm/VS Code
- Git/GitHub for version control

APPLICATIONS

- **E-Commerce Platforms –** try-on before buying.
- **Fashion Design & Retail –** digital clothing previews.
- **Personal Styling –** mix & match virtually.
- **Marketing & Ads –** interactive campaigns.
- **Education & Research –** AI in fashion tech.
- **Sustainability –** reduces physical samples & returns.

ADVANTAGES, LIMITATIONS & FUTURE SCOPE

ADVANTAGE	LIMITATION	FUTURE SCOPE
Realistic, photo-like results	Dependent on image quality	Real-time AR try-on
Low cost, open-source tools	Struggles with complex clothing	Personalized body models
Integrable with e-commerce	GPU needed for speed	Mobile app & VR integration

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THANK YOU